

Auteur: Mark Gijsbrechts
Studierichting: Informatica
Oefeningenreeks 2B nr. 1.8

Druk $\tan(\text{Bgc} \cos x)$ uit in x :

$$\tan(\text{Bgc} \cos x) = \frac{\sin(\text{Bgc} \cos x)}{\cos(\text{Bgc} \cos x)}$$

$$\tan(\text{Bgc} \cos x) = \frac{\sin(\text{Bgc} \cos x)}{x}$$

$$\alpha = \text{Bgc} \cos x \text{ met } \alpha \in [0, \pi]$$

$$\Leftrightarrow x = \cos \alpha$$

$$x^2 = \cos^2 \alpha$$

$$x^2 = 1 - \sin^2 \alpha$$

$$\sin^2 \alpha = 1 - x^2$$

$$\sin \alpha = \sqrt{1 - x^2}$$

$$\sin(\text{Bgc} \cos x) = \sqrt{1 - x^2}$$

$$\tan(\text{Bgc} \cos x) = \frac{\sqrt{1 - x^2}}{x}$$

References